

ASE LAB 8

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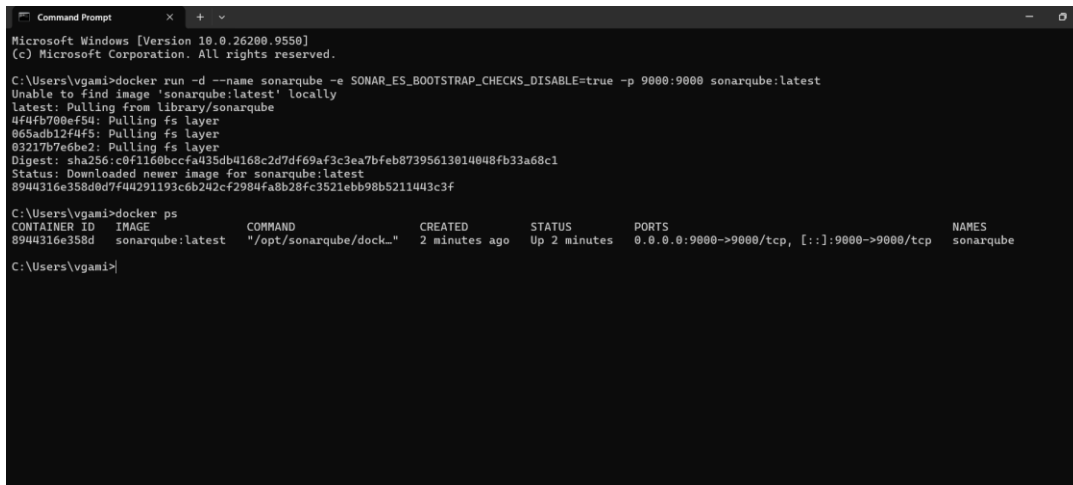
Adaptive Software Engineering – S6

Step 1:

- Firstly Open the docker desktop
- Then run this command:

```
docker run -d --name sonarqube -e  
SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000  
sonarqube:latest
```

- Wait until the command and then run: docker ps



```
Microsoft Windows [Version 10.0.26200.9550]
(c) Microsoft Corporation. All rights reserved.

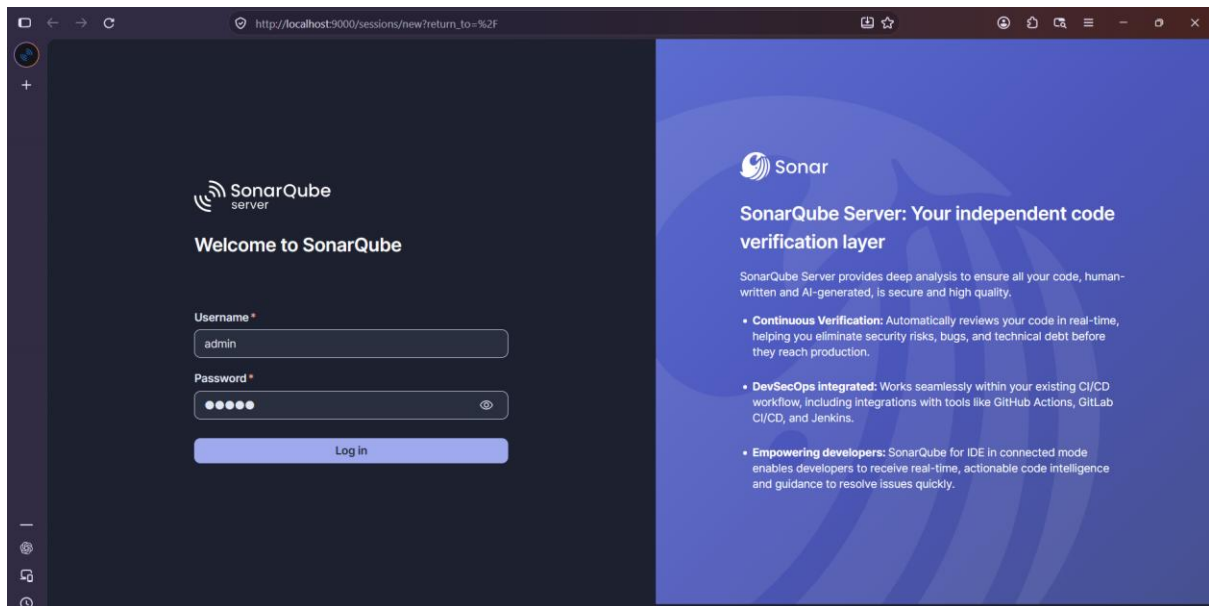
C:\Users\vgami>docker run -d --name sonarqube -e SONAR_ES_BOOTSTRAP_CHECKS_DISABLE=true -p 9000:9000 sonarqube:latest
Unable to find image 'sonarqube:latest' locally
latest: Pulling from library/sonarqube
4f4fb70ef54: Pulling fs layer
065adb12f4f5: Pulling fs layer
03217b7e6be2: Pulling fs layer
Digest: sha256:c0f1160bccfa425db4168c2d7df69af3c3ea7bfeb87395613014048fb33a68c1
Status: Downloaded newer image for sonarqube:latest
8944316e358d0d7f44291193c6b242cf2984fa8b28fc3521ebb98b5211443c3f

C:\Users\vgami>docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
8944316e358d   sonarqube:latest  "/opt/sonarqube/dock..."  2 minutes ago  Up 2 minutes  0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp  sonarqube

C:\Users\vgami>
```

Step 2: Open sonarqube using localhost

<http://localhost:9000>



➔ Then login using the credentials:

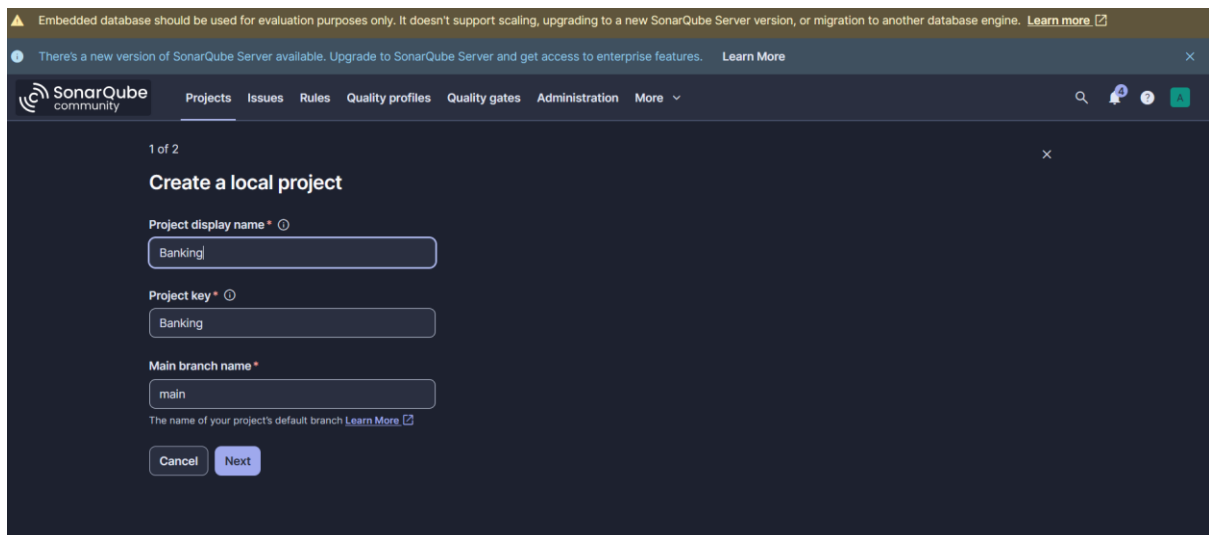
Username: admin

Password: admin

Step 3: Create the SonarQube project

Project Name: Banking

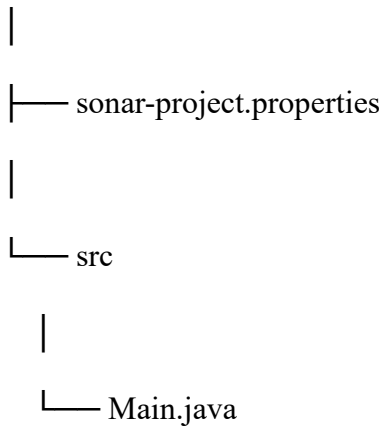
Project Key: Banking



STEP 4 — Create your Java project folder

The structure should be like this:

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sonar-project.properties:

sonar.projectKey=Banking

sonar.projectName=Banking

sonar.sources=src

sonar.host.url=http://localhost:9000

Main.java:

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        int a = 10;
```

```
        int b = 20;
```

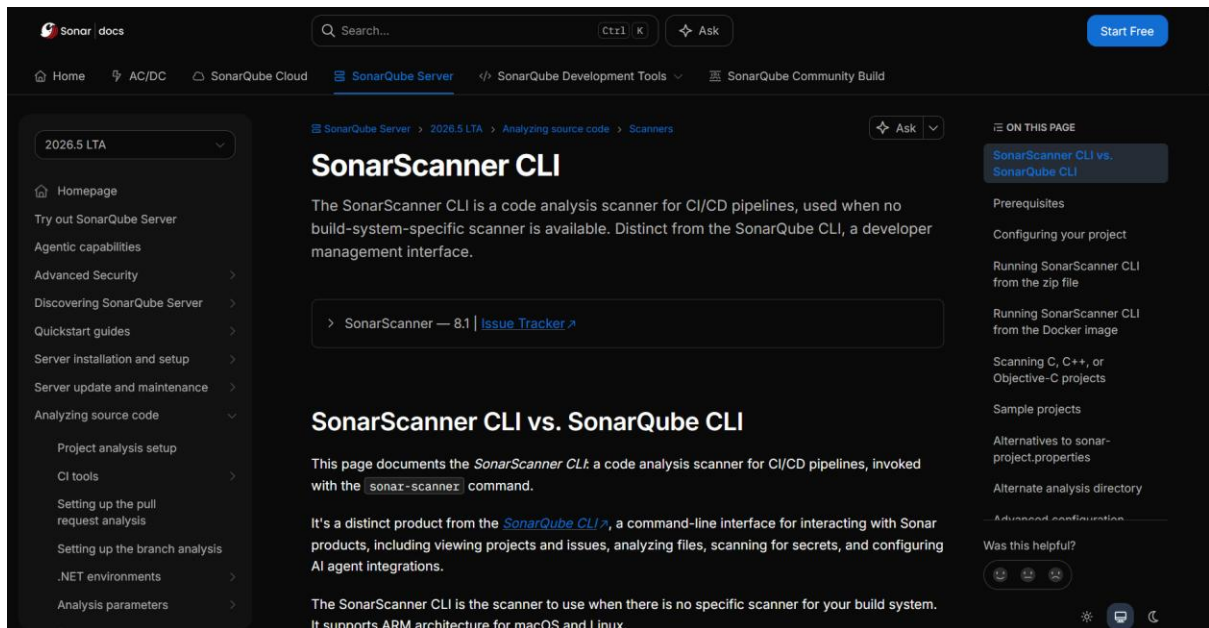
```
        System.out.println("Result: " + calculate(a, b));
```

```
        String name = null;
```

```
if (name != null) {  
    System.out.println(name);  
}  
  
int unused = 100;  
  
for (int i = 0; i < 10; i++) {  
    System.out.println(i);  
}  
}  
  
public static int calculate(int a, int b) {  
    return a + b;  
}  
  
public static void badMethod() {  
    int x = 10;  
    int y = 20;  
    int z = x + y;  
  
    System.out.println("This method is not needed");  
}  
}
```

Step 5: Install SonarScanner

<https://docs.sonarsource.com/sonarqube-server/analyzing-source-code/scanners/sonarscanner>

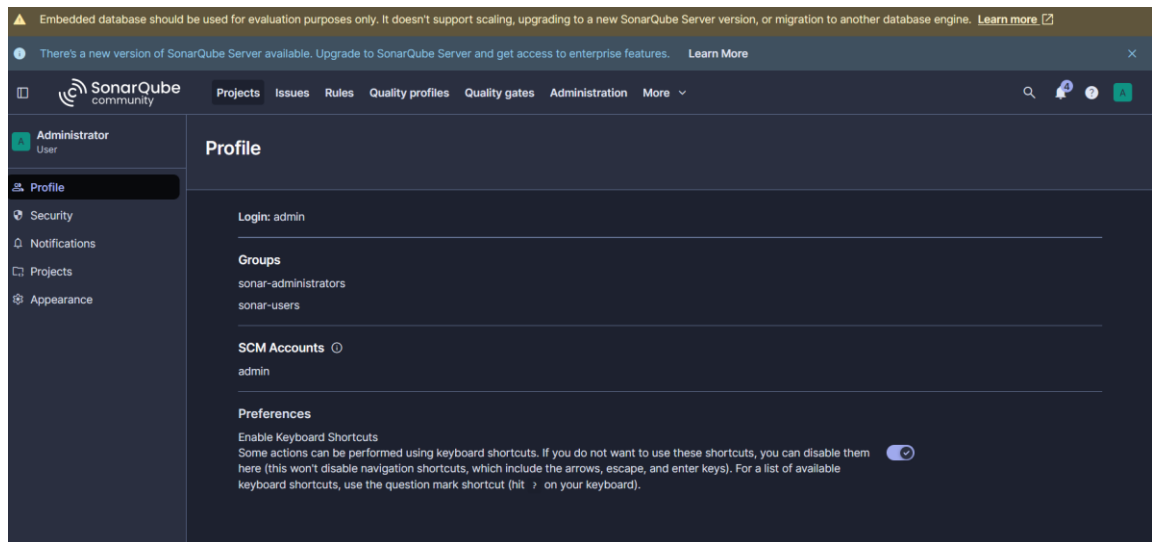


➔ To check if its installed correctly use: `sonar-scanner --version`

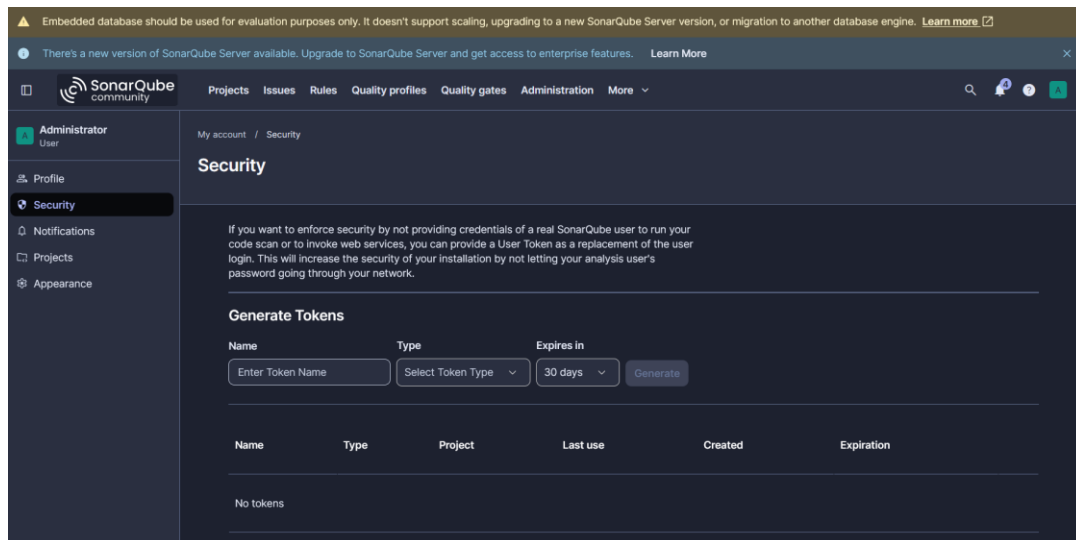
Step 6: Generate SonarQube authentication token

➔ Go to the localhost:9000, login using the credentials

➔ After logging in open your profile on the right corner



➔ Then go to the Security option in the left



- ➔ Give a name to the token and then select the type as User token and generate it
- ➔ Copy the token and open the cmd and run
set SONAR_TOKEN=YOUR_TOKEN

Step 7: Go to your project directory

- ➔ Open cmd and then go to your project directory ASE-8

Step 8: Run SonarScanner

- ➔ After going to your directory in cmd run the command:
sonar-scanner

```

C:\Users\vgami\OneDrive\Desktop\3rd Year\OOD\ASE\ASE-8>sonar-scanner
15:07:33.833 INFO Scanner configuration file: C:\Users\vgami\Downloads\sonar-scanner-cli-8.1.0.6389-windows-x64\sonar-scanner-8.1.0.6389-windows-x64\bin\..\conf\sonar-scanner.properties
15:07:33.836 INFO Project root configuration file: C:\Users\vgami\OneDrive\Desktop\3rd Year\OOD\ASE\ASE-8\sonar-project.properties
15:07:33.879 INFO SonarScanner CLI 8.1.0.6389
15:07:33.890 INFO Windows 11 10.0 amd64
15:07:39.957 INFO Communicating with SonarQube Community Build 26.9.0.129388
15:07:39.959 INFO JRE provisioning: os[windows], arch[amd64]
15:07:46.378 INFO Starting SonarScanner Engine...
15:07:46.379 INFO Java 21.0.11 Eclipse Adoptium (64-bit)
15:07:52.268 INFO Load global settings
15:07:52.532 INFO Load global settings (done) | time=269ms
15:07:52.538 INFO Server id: 147b11e-AaELV_oVCzG6DkaFzVml
15:07:52.558 INFO Loading required plugins
15:07:52.558 INFO Load plugins index
15:07:52.636 INFO Load plugins index (done) | time=77ms
15:07:52.637 INFO Load/download plugins
15:07:53.286 INFO Load/download plugins (done) | time=569ms
15:07:53.737 INFO Process project properties
15:07:53.758 INFO Process project properties (done) | time=21ms
15:07:53.782 INFO Project key: Banking
15:07:53.783 INFO Base dir: C:\Users\vgami\OneDrive\Desktop\3rd Year\OOD\ASE\ASE-8
15:07:53.783 INFO Working dir: C:\Users\vgami\OneDrive\Desktop\3rd Year\OOD\ASE\ASE-8\.scannerwork
15:07:53.794 INFO Load project settings for component key: 'Banking'
15:07:53.832 INFO Load project settings for component key: 'Banking' (done) | time=38ms
15:07:53.874 INFO Load quality profiles
15:07:54.169 INFO Load quality profiles (done) | time=295ms
15:07:54.218 WARN SCM provider autodetection failed. Please use 'sonar.scm.provider' to define SCM of your project, or disable the SCM Sensor in the project settings.
15:07:54.259 INFO Load active rules
15:07:55.232 INFO Load active rules (done) | time=974ms
15:07:55.253 INFO Load analysis cache
15:07:55.271 INFO Load analysis cache (usu) | time=19ms
15:07:55.476 INFO Preprocessing files...
15:07:55.515 INFO 1 language detected in 1 preprocessed file (done) | time=38ms
15:07:55.518 INFO Loading plugins for detected languages
15:07:55.519 INFO Load/download plugins
15:07:56.378 INFO Load/download plugins (done) | time=859ms
15:07:56.789 INFO Load project repositories
15:07:57.191 INFO Load project repositories (done) | time=482ms
15:07:57.264 INFO Indexing files...
15:07:57.265 INFO Project configuration:
15:07:57.286 INFO 1 file indexed (done) | time=21ms
15:07:57.288 INFO Quality profile for Java: Sonar way
15:07:57.289 INFO ----- Run sensors on module Banking
15:07:57.583 INFO Load metrics repository
15:07:57.588 INFO Load metrics repository (done) | time=46ms
15:07:59.287 INFO Sensor IAC VML File Type Classification Sensor [iac]
15:07:59.211 INFO Sensor IAC VML File Type Classification Sensor [iac] (done) | time=1ms
15:07:59.212 INFO Sensor JavaSensor [java]
15:07:59.242 INFO Server-side caching is enabled. The Java analyzer will not try to leverage data from a previous analysis.
15:07:59.288 INFO Using ECJ batch to parse 1 Main Java source files with batch size 312 KB

```

```

15:08:01.016 INFO There are no files to be analyzed for the Java language
15:08:01.017 INFO Sensor Java Config Sensor [iac] (done) | time=1ms
15:08:01.017 INFO Sensor IAC Docker Sensor [iac]
15:08:01.019 INFO There are no files to be analyzed for the Docker language
15:08:01.021 INFO Sensor IAC Docker Sensor [iac] (done) | time=4ms
15:08:01.021 INFO Sensor SurefireSensor [java]
15:08:01.022 INFO parsing [C:\Users\vgami\OneDrive\Desktop\3rd Year\OOD\ASE\ASE-8\target\surefire-reports]
15:08:01.023 INFO Sensor SurefireSensor [java] (done) | time=4ms
15:08:01.024 INFO Sensor TextAndSecretsSensor [text]
15:08:01.155 INFO Available processors: 16
15:08:01.158 INFO Using 16 threads for analysis.
15:08:01.519 INFO The property 'sonar.tests' is not set. To improve the analysis accuracy, we categorize a file as a test file if any of the following is true:
  * The filename starts with "test"
  * The filename contains "test" or "tests"
  * Any directory in the file path is named: "doc", "docs", "test", "tests", "mock" or "mocks"
  * Any directory in the file path has a name ending in "test" or "tests"
15:08:01.578 INFO Start fetching files for the text and secrets analysis
15:08:01.814 INFO Using Git CLI to retrieve dirty files
15:08:01.980 WARN Retrieving only language associated files, make sure to run the analysis inside a git repository to make use of inclusions specified via "sonar.text.inclusions"
15:08:01.981 INFO Starting the text and secrets analysis
15:08:01.988 INFO 1 source file to be analyzed for the text and secrets analysis
15:08:01.924 INFO 1/1 source file has been analyzed for the text and secrets analysis
15:08:01.928 INFO Sensor TextAndSecretsSensor [text] (done) | time=901ms
15:08:01.928 INFO ----- Run sensors on project
15:08:02.186 INFO Sensor JaCoCo Aggregate XML Report Importer [jacoco]
15:08:02.187 INFO Sensor JaCoCo Aggregate XML Report Importer [jacoco] (done) | time=8ms
15:08:02.189 INFO Sensor IAC Project Sensor [iac]
15:08:02.189 INFO Sensor IAC Project Sensor [iac] (done) | time=3ms
15:08:02.189 INFO Sensor JavaProjectSensor [java]
15:08:02.196 INFO Sensor JavaProjectSensor [java] (done) | time=6ms
15:08:02.196 INFO Sensor Zero Coverage Sensor
15:08:02.205 INFO Sensor Zero Coverage Sensor (done) | time=8ms
15:08:02.205 INFO Sensor Java CPD Block Indexer
15:08:02.223 INFO Sensor Java CPD Block Indexer (done) | time=18ms
15:08:02.224 INFO ----- Gather SCA dependencies on project
15:08:02.229 INFO Dependency analysis skipped
15:08:02.232 INFO SCM Publisher No SCM system was detected. You can use the 'sonar.scm.provider' property to explicitly specify it.
15:08:02.236 INFO CPD Executor Calculating CPD for 1 file
15:08:02.245 INFO CPD Executor CPD calculation finished (done) | time=6ms
15:08:02.407 INFO Analysis report generated in 154ms, dir size:294.9 kB
15:08:02.644 INFO Analysis report compressed in 236ms, zip size=32.6 kB
15:08:02.707 INFO Analysis report uploaded in 122ms
15:08:02.770 INFO ANALYSIS SUCCESSFUL, you can find the results at: http://localhost:9000/dashboard?id=Banking
15:08:02.770 INFO Note that you will be able to access the updated dashboard once the server has processed the submitted analysis report
15:08:02.773 INFO More about the report processing at http://localhost:9000/api/ce/task?id=74be2682-59c4-41bc-9fc8-65b5cbe158df
15:08:02.792 INFO Analysis total time: 9.401 s
15:08:02.857 INFO SonarScanner Engine completed successfully
15:08:02.922 INFO EXECUTION SUCCESS
15:08:02.923 INFO Total time: 29.095s

C:\Users\vgami\OneDrive\Desktop\3rd Year\OOD\ASE\ASE-8>

```

➔ When you see the execution as successful go the localhost:9000

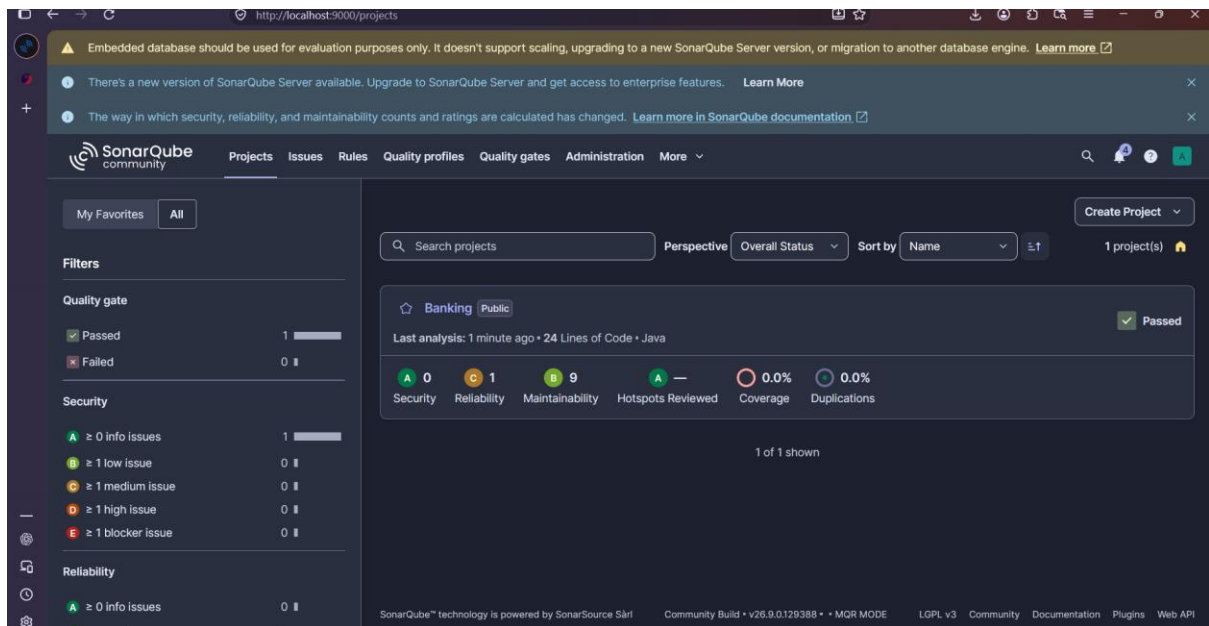
Step 9: Open your project dashboard

Go to:

<http://localhost:9000>

Then:

Projects → Banking



Here u should see your code if its passed or failed.